

5. The periodic table of elements is ordered by atomic number and electron configuration.

The tables below show some properties of alkali metals (group 1) and halogens (group 7).

Alkali metal	Atomic number	Melting point (°C)	Boiling point (°C)	Density at 20°C (g/cm ³)	Radius of the atom (units)
lithium	3	180.5	1342	0.534	1.67
sodium	11	97.7	883	0.971	
potassium	19		759	0.862	2.43
rubidium	37	39.3	688	1.532	2.65
caesium	55	28.4	671	1.873	2.98
francium	87	27.0	677		

Halogen	Atomic number	Melting point (°C)	Boiling point (°C)	Density at 20°C (g/cm ³)	Radius of the atom (units)
fluorine	9	−219	−188	0.0017	0.64
chlorine	17		−34	0.0032	0.99
bromine	35	−7	59	3.1	1.14
iodine	53	114	184	4.9	1.33
astatine	85	302	380	7.0	1.40

Use the information in the tables and your knowledge to answer the following questions.

(a) **Complete** the tables by estimating the missing values. [3]

(b) Compare the properties of the alkali metals and their trends with those of the halogens. [6 QER]

.....

.....

.....

.....

.....

.....

.....

(c) Explain why the reactivity of halogens changes from the top to the bottom of the group. [3]

(d) Tick (✓) the box next to the pair of elements that would react **most** violently. [1]

chlorine and lithium ☐

fluorine and lithium ☐

chlorine and sodium ☐

fluorine and sodium ☐